

Chrysomya bezziana

Status: Currently not present in Australia

Are you ready to submit maggot samples to Northern Australia Quarantine Strategy team for identification?

Profile

Screwworm fly is an insect parasite of warm-blooded animals (including people and birds). Flies can lay eggs in many types of wounds and moist areas on animals or people. Wounds can include small scratches, branding marks or castration wounds.

Once the eggs hatch the maggots feed on the flesh of their host. Unlike other flies, whose maggots can be found in dead animals, screwworm fly maggots only feed on tissue in live animals. The flies can multiply quickly – maggots can hatch from eggs within 24 hours in warm, tropical climates like that of northern Australia. The maggots can cause debilitating tissue damage, leading to significant production losses and possibly even death.

Identification

Infestation with *Chrysomya bezziana* (Old World screwworm fly) looks just like an Australian blowfly with a shiny blue-green body and a yellow face. Maggots may be hard to see, as they often burrow deep in the wound. A fly-struck wound will often weep and emit a pungent, rotten smell.

Distribution

Old World screwworm fly is found in parts of Africa, the Middle East, Asia and the Pacific, including the areas of Papua New Guinea that border Torres Strait.

A similar fly (New World screwworm fly) has been eradicated in the United States at great cost, but it is still found, and is spreading, in many countries in the Americas. Both flies would have the same impact if they became established in Australia. However, the Old World screwworm fly is considered a greater risk due to its proximity to Australia.

Threat

Animals infested with screwworm fly may eventually die from loss of tissue, fluid and infection.

If screwworm fly were to establish in Australia it would have disastrous effects on our:

- livestock industry (particularly cattle and sheep)
- native wildlife
- human health

Livestock industries could face millions of dollars a year in lost production and control measures. Once established in native wildlife, it would be very difficult to contain and eradicate.

Pathways

Screwworm fly can be carried in infested wounds on livestock, wildlife or humans moving from areas where it is present.

Potential risk pathways for screwworm fly to enter into Australia include:

- Movement of infested animals from countries where screwworm fly is present.
- Itinerant vessels such as yachts or illegal foreign fishing vessels carrying animals with infested wounds
- Travellers returning from countries where the pest is endemic may also be infested
- Wind patterns from the north could naturally disperse adult flies
- Returning livestock export vessels to northern ports

Due to close proximity to endemic populations of screwworm fly, Torres Strait is a vulnerable entry point along with areas in northern Australia at international ports.

What to look for

Check animals or humans for wounds and for signs of maggots deep within any wound. Wounds that can become infested include husbandry wounds (for example ear-tagging, castration and dehorning) and relatively minor skin wounds (such as tick bites). Maggots can also be found in other body areas, including the umbilicus, eyes, genital area and mouth

What you can do

Avoid moving infested animals from countries where screwworm fly is present.

Moving live animals, untanned hides or skins out of the Torres Strait Protected Zone to the Torres Strait Permanent Biosecurity Monitoring Zone, or from either zone to mainland Australia without a permit and an inspection by a departmental biosecurity officer is prohibited.

Record the time, date and location, if you see maggots in deep wounds and take photos.

Report:

- Maggots found in deep wounds of animals, birds or humans via the [Maggot Collection Form](#) for identification.
- Alternatively contact a ranger, biosecurity officer or government vet or call the [Emergency Animal Disease Hotline](#) on 1800 675 888.
- Incidents of animals being transported illegally to Australia from neighbouring countries.



Live animals and untreated hides or skins need a permit to move south between zones and to mainland Australia.

Keep a Top Watch!

Keep a Top Watch! for maggots found in deep wounds of live animals or humans.

If you see maggots in living tissue, collect and submit for identification:

- Take photos
- Record the time, date and location
- Collect 10 maggots
- Pour hot water over maggots. Allow to sit for 2 min.
(If hot water is not immediately available samples can be stored in fresh water for up to 24 hours).
- Drain water. Cover maggots in alcohol-based hand sanitiser or 70%+ ethanol. Seal container and place in zip-lock bag.
- Submit for identification via the [Maggot Collection Form](#).
(Link will take you to a form to collect core details and where to send maggots.)
- Alternatively call the [Emergency Animal Disease](#) Hotline on 1800 675 888 for details.

As part of the wider screwworm fly surveillance activity, the department distributes screwworm fly collection kits to pastoralists, health clinics, ranger groups and vets throughout northern Australia for the collection and identification of maggots.

For a free kit or for more information please contact [NAQS](#).

For more information in your jurisdiction go to:

- Northern Territory: [Screw-worm fly surveillance and preparedness program | NT.GOV.AU](#)
- Queensland: [Screw-worm fly | Business Queensland](#)
- Western Australia: [Department of Primary Industries and Regional Development \[DPIRD\]](#)
- Animal Health Australia: [Monitoring for screwworm fly](#)

Source: <https://www.agriculture.gov.au/biosecurity/australia/naqs/naqs-target-lists/screw-worm-fly>